Uncertainty tracking across material-flow interfaces

Proposed synthesis: reported capabilities mapped to explicit interface requirements

A. Intended use and evidence requirement

Quantity of interest | spatial support | time horizon | tolerance | decision authority

S1	S2	S3	S4	S5
Attributes	Inventory	Dated outflows	Recovery	Impacts

B. Shared interface contract — proposed cross-study connection

Entity + variable + unit | U representation + dependencies | transformation + assumptions
Spatial support + valid time | version + evidence | downstream use

Observe	Infer	Propagate	Diagnose	Decide
Quality / timing Missing or stale data	State / parameter U Supported uncertainty	Samples / dependencies Input-to-output mapping	Residuals / coverage Mismatch, not unique cause	Risk / scenario context Conditions and authority

C. Traceable update and explanation

Retain previous state → record new evidence → revise the relevant model → reassess the decision
Capability concepts: R2, R3, R5, R6, R15; allocation to all S1–S5 boundaries is our synthesis.

Diagnostic limits

A residual can flag disagreement; it does not identify every uncertainty source.

No universal threshold, automatic reduction or end-to-end deployment is claimed.